Baligen Talihati

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Education

University of Tennessee, Knoxville **Aug. 2025 – Present**
Ph.D. *Electrical Engineering and Computer Science*
Advisor: Hao Huang
Research Area: Power Grid Resilience.

Fudan University **Sept. 2022 – Jun. 2025**
M.S. *Electronics Science and Technology, GPA 3.71/4.00*
Core Modules: Moden Control Theory and Technology, Modern Sensor Technology, Detection of Weak Signal.
Thesis: Community Energy Management Method Based on Multi-Agent Reinforcement Learning.

Fudan University **Sept. 2016 – Jun. 2020**
B.E. *Electrical Engineering and Automation, GPA 3.16/4.00*
Core Modules: Mathematical Analysis, C Programming Language, University Physics, Linear Algebra, Automatic Control Technology.

Research Interests

Smart Grid, Renewable Energy, Machine Learning

Publications and Patents

Publications:

[J.1] Baligen Talihati, Shengyu Tao, Shiyi Fu, Bowen Zhang, Hongtao Fan, Qifen Li, Xiaodong Lv, Yaojie Sun, Yu Wang, Energy storage sharing in residential communities with controllable loads for enhanced operational efficiency and profitability, ***Applied Energy*** 373 (2024) 123880. doi:10.1016/j.apenergy.2024.123880. (JCR Q1 IF=11.0, Accepted on July 9, 2024)

[J.2] Baligen Talihati, Shiyi Fu, Bowen Zhang, Yuqing Zhao, Yu Wang, Yaojie Sun, Community shared ES-PV system for managing electric vehicle loads via multi-agent reinforcement learning, ***Applied Energy*** 380 (2025) 125039. doi: 10.1016/j.apenergy.2024.125039. (JCR Q1 IF=11.0, Accepted on November 25, 2024)

[J.3] Bowen Zhang, **Baligen Talihati**, Hongtao Fan, Yaojie Sun, Yu Wang, A dynamic carbon flow traceability framework for integrated energy systems, ***Journal of Cleaner Production*** 518 (2025) 145878. doi: 10.1016/j.jclepro.2025.145878. (JCR Q1 IF=10.0, Accepted on May 31, 2025)

Patents:

[P.1] Yu Wang, **Baligen Talihati**, Yaojie Sun, Bowen Zhang. Community energy storage sharing framework with power dispatch method. China National Intellectual Property Administration (CNIPA). Patent No.20230036. Filed September 6, 2023. Patent application accepted.

[P.2] Yaojie Sun, Yuanhui Chen, Yu Wang, **Baligen Talihati**, Bowen Zhang. Shared energy storage - multi-producer and consumer cooperative optimization method. China National Intellectual Property Administration (CNIPA). Patent No.20230003. Filed Febrary 10, 2023. Patent application accepted.

Work Experience

Substation Secondary Protection Maintenance, State Grid **July. 2020 – Oct. 2021**
Urumqi Power Supply Company *Urumqi, CHN*

- Responsible for the inspection and routine maintenance of secondary protection devices at 13 substations in Urumqi.
- Conducted regular equipment inspections and testing to ensure operational reliability and compliance with safety standards; participated in scheduled maintenance and emergency response activities.

- Collaborated with primary equipment engineers on substation equipment commissioning, system integration testing, and coordination of protection device settings during equipment upgrades and modifications.
- **Technical Proficiency**: Developed strong expertise in control circuits for secondary protection systems, DC power systems, and SCADA communication protocols; gained hands-on experience with relay testing equipment and diagnostic tools.

Industrial AI Algorithm Application Intern, Siemens — Sept. 2024 – Mar.2025

Siemens Digital R&D Center — *Shanghai, CHN*

- Acquired proficiency in AI algorithms for industrial equipment anomaly detection, enhancing understanding of machine learning principles, types of industrial anomalies, and their root causes.
- Performed data processing tasks for equipment anomaly data, ensuring accurate analysis and timely insights for preventive measures.
- **Technical Proficiency**: Developed expertise in Python and scikit-learn libraries, utilizing advanced algorithms to improve system diagnostics and operational efficiency.

Research Experience

Community Energy Management Systems — Sept. 2023 – June. 2024

Graduate Researcher, Artificial Intelligence Energy Control Laboratory, Fudan University — *Shanghai, CHN*

- Prof. Yaojie Sun, Supervisor
- Developed a Community Energy Management System model using Python and Pytorch, based on Multi-Agent Reinforcement Learning.
- Evaluated effectiveness using metrics such as solar self-consumption and cost reduction.
- Enhanced solar self-consumption by 66.41% and reduced electricity costs by 7.73%, supporting a total load capacity of 38.68%.

Carbon Neutral Project in Industrial Park — Dec. 2022 – June. 2024

Graduate Researcher, State Grid Jiangsu Company — *Nanjing, CHN*

- Prof. Yaojie Sun, Supervisor
- Developed microgrid models under different cooperative storage modes.
- Assisted in a 100-million RMB project led by State Grid, ensuring smooth progress and timely delivery.
- Implemented shared storage-based systems at a factory site, assessing carbon footprint to support carbon neutrality.

Fellowships and Awards

Shanghai Municipal Outstanding Graduate Award

Awarded for outstanding research performance — *Sept. 2022 – Apr. 2025*

Corporate-Sponsored Scholarship, approximately $1100

Awarded for outstanding research performance — *Sept. 2023 – Sept. 2024*

Outstanding Student Scholarship, approximately $1200

Awarded for outstanding academic performance — *Sept. 2018 – Jun. 2019*

Teaching Assistant

University of Tennessee, Knoxville

- *ECE525 Alternative Energy Source* — *Fall 2025*
- *ECE581 High Frequency Power Electronics* — *Fall 2025*

Fudan University

- *INFO130263.01 Electronic Methods* — *Spring 2023*

Skills & Languages

Proficient in Coding: Python, MATLAB, LaTeX.

Languages: Mandarin(Native), Kazakh Language(Native), English(IELTS: Overall 7.0)